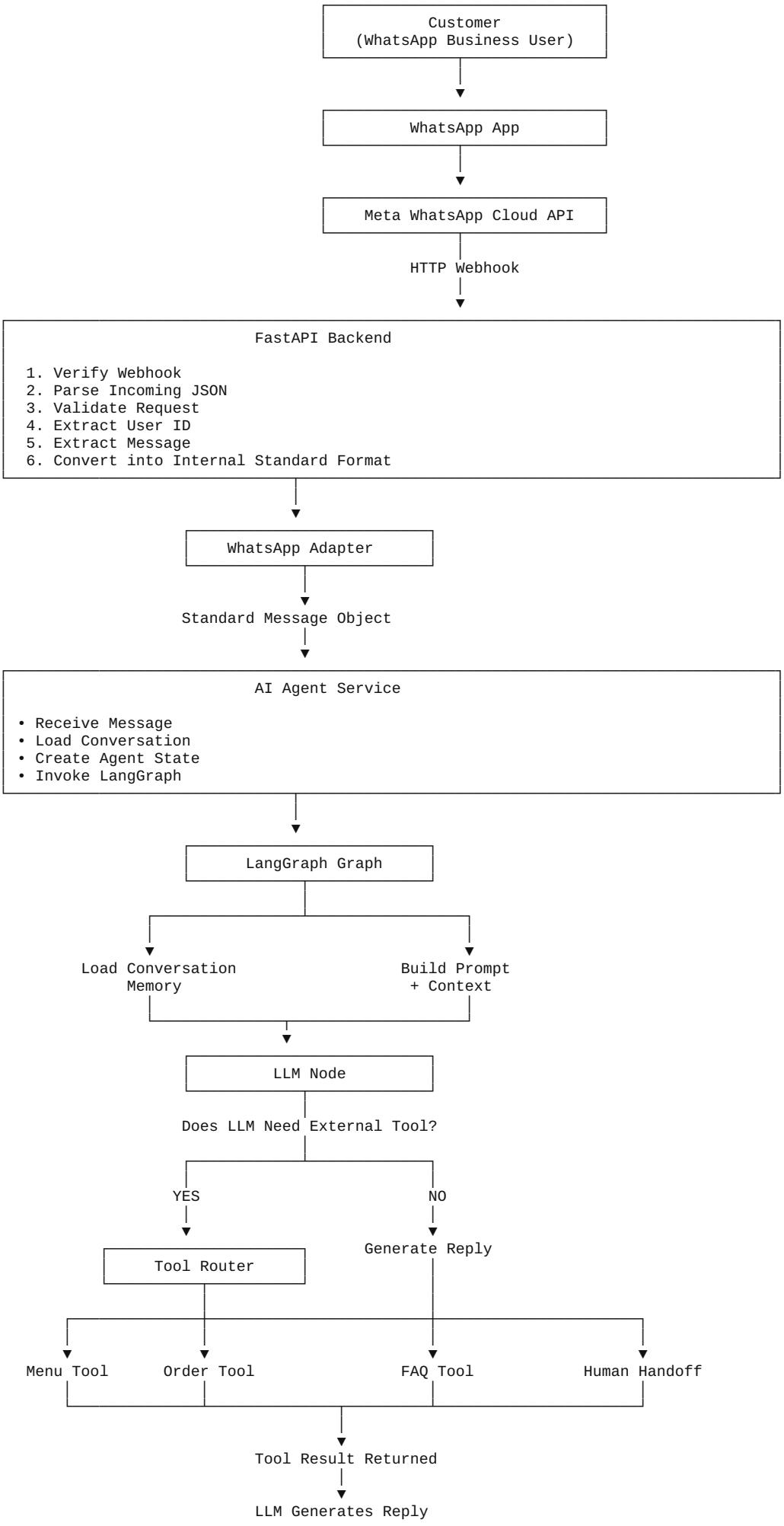
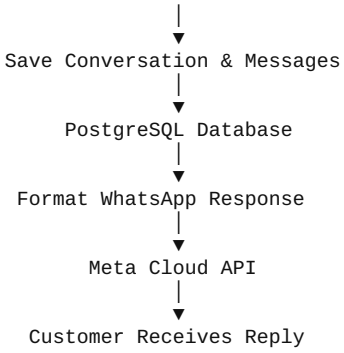

Customer
(WhatsApp Business User)
WhatsApp App
Meta WhatsApp Cloud API
HTTP Webhook
FastAPI Backend
1. Verify Webhook
2. Parse Incoming JSON
3. Validate Request
4. Extract User ID
5. Extract Message
6. Convert into Internal Standard Format
WhatsApp Adapter
Standard Message Object
AI Agent Service
• Receive Message
• Load Conversation
• Create Agent State
• Invoke LangGraph
LangGraph Graph
Load Conversation Memory
Build Prompt + Context
LLM Node
Does LLM Need External Tool?
YES
NO
Tool Router
Generate Reply
Menu Tool
Order Tool
FAQ Tool
Human Handoff
Tool Result Returned
LLM Generates Reply


Save Conversation & Messages
PostgreSQL Database
Format WhatsApp Response
Meta Cloud API
Customer Receives Reply